

SPACE PROGRAMME: ISRO, ORGANISATION AND LAUNCH VEHICLES

INSTITUTIONAL COMMAND MAP → CENTRE-TO-LAUNCH CHAIN → POLICY AND LAW STACK → LAUNCHER-PAYLOAD SEPARATION → ORBIT VOCABULARY MATRIX → GTO-TO-GEO RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 INSTITUTIONAL COMMAND MAP

INSTITUTIONAL COMMAND MAP DEPARTMENT OF SPACE POLICY UMBRELLA AND SOVEREIGN DIRECTION PUBLIC R&D, LAUNCH VEHICLES, SPACECRAFT AND MISSIONS PROMOTE, AUTHORISE AND SUPERVISE NON-GOVERNMENT ACTIVITY COMMERCIAL CONTRACTS, SERVICES AND TECHNOLOGY TRANSFER

INSTITUTION NAMES ARE NOT INTERCHANGEABLE

DEPARTMENT OF SPACE → policy umbrella and sovereign direction

ISRO → public R&D, launch vehicles, spacecraft and missions

IN-SPACe → promote, authorise and supervise non-government activity

NSIL → commercial contracts, services and technology transfer

CONCEPT / ARCHITECTURE

- DEPARTMENT OF SPACE → policy umbrella and sovereign direction
- ISRO → public R&D, launch vehicles, spacecraft and missions

MECHANISM / APPLICATION

- IN-SPACe → promote, authorise and supervise non-government activity
- NSIL → commercial contracts, services and technology transfer

READINESS / STATUS LIMIT

- RULE → institution names are not interchangeable
- DEPARTMENT OF SPACE → policy umbrella and sovereign direction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NSIL → commercial contracts, services and technology transfer.

01

STAGE 01 CENTRE-TO-LAUNCH CHAIN

CENTRE-TO-LAUNCH CHAIN LAUNCH-VEHICLE DESIGN AND DEVELOPMENT LIQUID PROPULSION AND PROPULSION TESTING ROLES INTEGRATION, RANGE AND LAUNCH OPERATIONS MISSION CONTROL FLIGHT MONITORING AND DECISION CHAIN A CENTRE FUNCTION IS NOT THE WHOLE ISRO MANDATE

VSSC → launch-vehicle design and development

LPSC / IPRC → liquid propulsion and propulsion testing roles

SDSC-SHAR → integration, range and launch operations

MISSION CONTROL → flight monitoring and decision chain

RULE → a centre function is not the whole ISRO mandate

STARTING CONDITION / INPUT

- VSSC → launch-vehicle design and development
- LPSC / IPRC → liquid propulsion and propulsion testing roles

SCIENTIFIC OR ENGINEERING MECHANISM

- SDSC-SHAR → integration, range and launch operations
- MISSION CONTROL → flight monitoring and decision chain

OUTPUT / FAILURE BOUNDARY

- RULE → a centre function is not the whole ISRO mandate
- VSSC → launch-vehicle design and development

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a centre function is not the whole ISRO mandate.

02

STAGE 02 POLICY AND LAW STACK

POLICY AND LAW STACK OUTER-SPACE TREATIES INTERNATIONAL STATE OBLIGATIONS INDIAN SPACE POLICY 2023 EXECUTIVE ROLE ALLOCATION IN-SPACE NORMS AUTHORISATION AND SUPERVISION INTERFACE COMMERCIAL CONTRACT

OUTER-SPACE TREATIES → international State obligations

INDIAN SPACE POLICY 2023 → executive role allocation

IN-SPACe NORMS → authorisation and supervision interface

COMMERCIAL CONTRACT → service and risk allocation

CONCEPT / ARCHITECTURE

- OUTER-SPACE TREATIES → international State obligations
- INDIAN SPACE POLICY 2023 → executive role allocation

MECHANISM / APPLICATION

- IN-SPACe NORMS → authorisation and supervision interface
- COMMERCIAL CONTRACT → service and risk allocation

READINESS / STATUS LIMIT

- GAP → policy is not a comprehensive domestic space statute
- OUTER-SPACE TREATIES → international State obligations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GAP → policy is not a comprehensive domestic space statute.

03

STAGE 03 LAUNCHER-PAYLOAD SEPARATION

LAUNCHER-PAYLOAD SEPARATION LAUNCH VEHICLE SUPPLIES ASCENT ENERGY AND GUIDANCE PAYLOAD FAIRING PROTECTS THE MISSION PAYLOAD LAUNCHER TASK AND SPACECRAFT TASK DIVERGE CARRIES PAYLOAD AND MAY RAISE OR MAINTAIN ORBIT

ROCKET SUCCESS AND SPACECRAFT SERVICE ENTRY ARE SEPARATE

LAUNCH VEHICLE → supplies ascent energy and guidance

PAYLOAD FAIRING → protects the mission payload

SEPARATION → launcher task and spacecraft task diverge

SPACECRAFT → carries payload and may raise or maintain orbit

RULE → rocket success and spacecraft service entry are separate

03

CONCEPT / ARCHITECTURE

- OUTER-SPACE TREATIES → international State obligations
- INDIAN SPACE POLICY 2023 → executive role allocation

MECHANISM / APPLICATION

- IN-SPACE NORMS → authorisation and supervision interface
- COMMERCIAL CONTRACT → service and risk allocation

READINESS / STATUS LIMIT

- GAP → policy is not a comprehensive domestic space statute
- OUTER-SPACE TREATIES → international State obligations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GAP → policy is not a comprehensive domestic space statute.

04

STAGE 03

LAUNCHER-PAYLOAD SEPARATION

LAUNCHER-PAYLOAD SEPARATION

LAUNCH VEHICLE

SUPPLIES ASCENT ENERGY AND GUIDANCE

PAYLOAD FAIRING

PROTECTS THE MISSION PAYLOAD

LAUNCHER TASK AND SPACECRAFT TASK DIVERGE

CARRIES PAYLOAD AND MAY RAISE OR MAINTAIN ORBIT

ROCKET SUCCESS AND SPACECRAFT SERVICE ENTRY ARE SEPARATE

LAUNCH VEHICLE → supplies ascent energy and guidance

PAYLOAD FAIRING → protects the mission payload

SEPARATION → launcher task and spacecraft task diverge

SPACECRAFT → carries payload and may raise or maintain orbit

RULE → rocket success and spacecraft service entry are separate

STARTING CONDITION / INPUT

LAUNCH VEHICLE → supplies ascent energy and guidance

PAYLOAD FAIRING → protects the mission payload

SCIENTIFIC OR ENGINEERING MECHANISM

SEPARATION → launcher task and spacecraft task diverge

SPACECRAFT → carries payload and may raise or maintain orbit

OUTPUT / FAILURE BOUNDARY

RULE → rocket success and spacecraft service entry are separate

LAUNCH VEHICLE → supplies ascent energy and guidance

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → rocket success and spacecraft service entry are separate.

05

STAGE 04

ORBIT VOCABULARY MATRIX

ORBIT VOCABULARY MATRIX

ALTITUDE CLASS, OFTEN USED BY CREW AND EARTH OBSERVATION

HIGH-INCLINATION GROUND-TRACK GEOMETRY

REPEATED LOCAL-SOLAR-TIME CROSSING

ORBITAL PERIOD TIED TO EARTH'S ROTATION

CIRCULAR EQUATORIAL GEOSYNCHRONOUS SPECIAL CASE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
LEO → altitude class, often used by crew and Earth observation	SUN-SYNCHRONOUS → repeated local-solar-time crossing	GEO → circular equatorial geosynchronous special case
POLAR → high-inclination ground-track geometry	GSO → orbital period tied to Earth's rotation	LEO → altitude class, often used by crew and Earth observation

SYSTEM / DEVICE / INSTITUTION

LEO → altitude class, often used by crew and Earth observation

POLAR → high-inclination ground-track geometry

DECISIVE TECHNICAL DISTINCTION

SUN-SYNCHRONOUS → repeated local-solar-time crossing

GSO → orbital period tied to Earth's rotation

STATUS / UNIT / EVIDENCE LIMIT

GEO → circular equatorial geosynchronous special case

LEO → altitude class, often used by crew and Earth observation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GSO → orbital period tied to Earth's rotation.

06

STAGE 05

GTO-TO-GEO RAIL

GTO-TO-GEO RAIL

LAUNCHER ASCENT

REACHES TRANSFER CONDITIONS

GTO INJECTION

ELLIPTICAL INTERMEDIATE ORBIT

SPACECRAFT PROPULSION

RAISES PERIGEE AND CIRCULARISES

GEO SLOT

LAUNCHER ASCENT → reaches transfer conditions

GTO INJECTION → elliptical intermediate orbit

SPACECRAFT PROPULSION → raises perigee and circularises

GEO SLOT → circular equatorial final service orbit

TRAP → GTO delivery is not direct GEO placement

STARTING CONDITION / INPUT

LAUNCHER ASCENT → reaches transfer conditions

GTO INJECTION → elliptical intermediate orbit

SCIENTIFIC OR ENGINEERING MECHANISM

SPACECRAFT PROPULSION → raises perigee and circularises

GEO SLOT → circular equatorial final service orbit

OUTPUT / FAILURE BOUNDARY

TRAP → GTO delivery is not direct GEO placement

LAUNCHER ASCENT → reaches transfer conditions

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → GTO delivery is not direct GEO placement.

06

STAGE 06

OPERATIONAL LAUNCHER MATRIX

OPERATIONAL LAUNCHER MATRIX

VERSATILE WORKHORSE AND MIXED MISSION PROFILES

CRYOGENIC UPPER STAGE AND TRANSFER-ORBIT ROLE

HEAVIER OPERATIONAL CLASS AND HUMAN-RATING BASE

SEPARATE SMALL-SATELLITE LAUNCH ARCHITECTURE

ROLE, PROPULSION, ORBIT AND STATUS MUST TRAVEL TOGETHER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PSLV → versatile workhorse and mixed mission profiles	LVM3 → heavier operational class and human-rating base	RULE → role, propulsion, orbit and status must travel together
GSLV → cryogenic upper stage and transfer-orbit role	SSLV → separate small-satellite launch architecture	PSLV → versatile workhorse and mixed mission profiles

SYSTEM / DEVICE / INSTITUTION

PSLV → versatile workhorse and mixed mission profiles

DECISIVE TECHNICAL DISTINCTION

LVM3 → heavier operational class and human-rating base

STATUS / UNIT / EVIDENCE LIMIT

RULE → role, propulsion, orbit and status must travel together

SPACE PROGRAMME: ISRO, ORGANISATION AND LAUNCH VEHICLES • TILE 2/4 • same master rows 2475-5709 of 10660 • continues below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → GTO delivery is not direct GEO placement.

06

STAGE 06 OPERATIONAL LAUNCHER MATRIX

OPERATIONAL LAUNCHER MATRIX

VERSATILE WORKHORSE AND MIXED MISSION PROFILES

CRYOGENIC UPPER STAGE AND TRANSFER-ORBIT ROLE

HEAVIER OPERATIONAL CLASS AND HUMAN-RATING BASE

SEPARATE SMALL-SATELLITE LAUNCH ARCHITECTURE

ROLE, PROPULSION, ORBIT AND STATUS MUST TRAVEL TOGETHER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PSLV → versatile workhorse and mixed mission profiles	LVM3 → heavier operational class and human-rating base	RULE → role, propulsion, orbit and status must travel together
GSLV → cryogenic upper stage and transfer-orbit role	SSLV → separate small-satellite launch architecture	PSLV → versatile workhorse and mixed mission profiles
SYSTEM / DEVICE / INSTITUTION • PSLV → versatile workhorse and mixed mission profiles • GSLV → cryogenic upper stage and transfer-orbit role	DECISIVE TECHNICAL DISTINCTION • LVM3 → heavier operational class and human-rating base • SSLV → separate small-satellite launch architecture	STATUS / UNIT / EVIDENCE LIMIT • RULE → role, propulsion, orbit and status must travel together • PSLV → versatile workhorse and mixed mission profiles

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → role, propulsion, orbit and status must travel together.

07

STAGE 07 LINEAGE AND STATUS LADDER

LINEAGE AND STATUS LADDER

EXPERIMENTAL ORBITAL-ACCESS LEARNING

INTERMEDIATE TECHNOLOGY-LEARNING STEP

OPERATIONAL FAMILIES IN OWNER

DEVELOPED SMALL-LAUNCH FAMILY WITH DATED STATUS NEEDED

APPROVED FUTURE DEVELOPMENT, NOT AN OPERATIONAL LAUNCHER



STARTING CONDITION / INPUT

- SLV-3 → experimental orbital-access learning
- ASLV → intermediate technology-learning step

SCIENTIFIC OR ENGINEERING MECHANISM

- PSLV / GSLV / LVM3 → operational families in owner
- SSLV → developed small-launch family with dated status needed

OUTPUT / FAILURE BOUNDARY

- NGLV → approved future development, not an operational launcher
- SLV-3 → experimental orbital-access learning

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NGLV → approved future development, not an operational launcher.

08

STAGE 08 PROPULSION COMPARISON

PROPULSION COMPARISON

STORED PROPELLANT AND LIMITED THROTTLE CONTROL

EARTH-STORABLE LIQUID

AMBIENT-TEMPERATURE LIQUID PAIR

LIQUID OXYGEN PLUS LIQUID HYDROGEN

LIQUID OXYGEN PLUS KEROSENE

PROPELLANT CLASS DOES NOT PROVE FLIGHT STATUS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SOLID → stored propellant and limited throttle control	CRYOGENIC → liquid oxygen plus liquid hydrogen	RULE → propellant class does not prove flight status
EARTH-STORABLE LIQUID → ambient-temperature liquid pair	SEMI-CRYOGENIC → liquid oxygen plus kerosene	SOLID → stored propellant and limited throttle control
SYSTEM / DEVICE / INSTITUTION • SOLID → stored propellant and limited throttle control • EARTH-STORABLE LIQUID → ambient-temperature liquid pair	DECISIVE TECHNICAL DISTINCTION • CRYOGENIC → liquid oxygen plus liquid hydrogen • SEMI-CRYOGENIC → liquid oxygen plus kerosene	STATUS / UNIT / EVIDENCE LIMIT • RULE → propellant class does not prove flight status • SOLID → stored propellant and limited throttle control

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → propellant class does not prove flight status.

09

STAGE 09 HUMAN-RATING GATE

HUMAN-RATING GATE

RELIABLE LAUNCHER

NECESSARY BUT INSUFFICIENT BASE

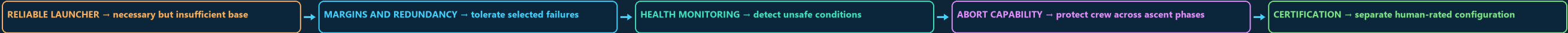
MARGINS AND REDUNDANCY

TOLERATE SELECTED FAILURES

HEALTH MONITORING

DETECT UNSAFE CONDITIONS

ABORT CAPABILITY



STARTING CONDITION / INPUT

- RELIABLE LAUNCHER → necessary but insufficient base
- MARGINS AND REDUNDANCY → tolerate selected failures

SCIENTIFIC OR ENGINEERING MECHANISM

- HEALTH MONITORING → detect unsafe conditions
- ABORT CAPABILITY → protect crew across ascent phases

OUTPUT / FAILURE BOUNDARY

- CERTIFICATION → separate human-rated configuration
- RELIABLE LAUNCHER → necessary but insufficient base

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ABORT CAPABILITY → protect crew across ascent phases.

09

STAGE 09

HUMAN-RATING GATE

HUMAN-RATING GATE

RELIABLE LAUNCHER

NECESSARY BUT INSUFFICIENT BASE

MARGINS AND REDUNDANCY

TOLERATE SELECTED FAILURES

HEALTH MONITORING

DETECT UNSAFE CONDITIONS

ABORT CAPABILITY

RELIABLE LAUNCHER → necessary but insufficient base

MARGINS AND REDUNDANCY → tolerate selected failures

HEALTH MONITORING → detect unsafe conditions

ABORT CAPABILITY → protect crew across ascent phases

CERTIFICATION → separate human-rated configuration

STARTING CONDITION / INPUT

- RELIABLE LAUNCHER → necessary but insufficient base
- MARGINS AND REDUNDANCY → tolerate selected failures

SCIENTIFIC OR ENGINEERING MECHANISM

- HEALTH MONITORING → detect unsafe conditions
- ABORT CAPABILITY → protect crew across ascent phases

OUTPUT / FAILURE BOUNDARY

- CERTIFICATION → separate human-rated configuration
- RELIABLE LAUNCHER → necessary but insufficient base

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ABORT CAPABILITY → protect crew across ascent phases.

10

STAGE 10

ACCESS-TO-SPACE SYSTEM

ACCESS-TO-SPACE SYSTEM

PROPULSION, STRUCTURE, AVIONICS AND GUIDANCE

REPEATABLE STAGE AND SUBSYSTEM QUALITY

INTEGRATION, SAFETY AND LAUNCH WINDOW

MISSION ASSURANCE

VERIFICATION AND ANOMALY CLOSURE

CADENCE AND CUSTOMER CONFIDENCE

VEHICLE → propulsion, structure, avionics and guidance

FACTORY → repeatable stage and subsystem quality

PAD / RANGE → integration, safety and launch window

MISSION ASSURANCE → verification and anomaly closure

CONCEPT / ARCHITECTURE

- VEHICLE → propulsion, structure, avionics and guidance
- FACTORY → repeatable stage and subsystem quality

MECHANISM / APPLICATION

- PAD / RANGE → integration, safety and launch window
- MISSION ASSURANCE → verification and anomaly closure

READINESS / STATUS LIMIT

- OUTPUT → cadence and customer confidence
- VEHICLE → propulsion, structure, avionics and guidance

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PAD / RANGE → integration, safety and launch window.

11

STAGE 11

FUTURE-CLAIM FIREWALL

FUTURE-CLAIM FIREWALL

INSTITUTION, VEHICLE, SPACECRAFT, ORBIT AND SERVICE

RETIRED, OPERATIONAL, APPROVED OR UNDER CONSTRUCTION

PROPULSION, INFRASTRUCTURE, INTEGRATION AND ASSURANCE

TECHNOLOGY TRANSFER FROM AUTHORISATION AND SAFETY

DATED OFFICIAL CAPACITY, SCHEDULE AND OUTCOME SOURCE

DEFINE → institution, vehicle, spacecraft, orbit and service

CLASSIFY → retired, operational, approved or under construction

TRACE → propulsion, infrastructure, integration and assurance

SEPARATE → technology transfer from authorisation and safety

VERIFY → dated official capacity, schedule and outcome source

SYSTEM / DEVICE / INSTITUTION

- DEFINE → institution, vehicle, spacecraft, orbit and service
- CLASSIFY → retired, operational, approved or under construction

DECISIVE TECHNICAL DISTINCTION

- TRACE → propulsion, infrastructure, integration and assurance
- SEPARATE → technology transfer from authorisation and safety

STATUS / UNIT / EVIDENCE LIMIT

- VERIFY → dated official capacity, schedule and outcome source
- DEFINE → institution, vehicle, spacecraft, orbit and service

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated official capacity, schedule and outcome source.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

STRATEGIC R&D ANCHOR AND TECHNOLOGY OWNER FOR LAUNCH-VEHICLE FAMILIES

SINGLE-WINDOW NODAL MECHANISM FOR PROMOTING, AUTHORISING AND SUPERVISING SPACE

COMMERCIAL PSU HANDLING LAUNCH SERVICES, SATELLITE SERVICES AND INDUSTRY-LINKED

SOVEREIGN POLICY UMBRELLA THAT STILL KEEPS SPACE AS A

THE REFORM QUESTION IS NOT "STATE VERSUS PRIVATE" BUT

REMAIN SOVEREIGN, WHICH CAN BE COMMERCIALY SCALED, AND HOW

LAUNCH AUTHORISATION INVOLVES LIABILITY, RANGE SAFETY, EXPORT-CONTROL SENSITIVITY, SPECTRUM/ORBITAL

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- ISRO: strategic R&D anchor and technology owner for launch-vehicle families and major missions.
- IN-SPACE: single-window nodal mechanism for promoting, authorising and supervising space activities of non-government entities.
- NSIL: commercial PSU handling launch services, satellite services and industry-linked production/commercial arrangements.
- DoS: sovereign policy umbrella that still keeps space as a strategic national capability domain.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: The reform question is not "state versus private" but "which function must remain sovereign, which can be commercially scaled, and how risk is supervised."
- CAUTION: Launch authorisation involves liability, range safety, export-control sensitivity, spectrum/orbital coordination and national-security oversight
- it cannot be treated like ordinary industrial licensing.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- CAUTION: Over-centralisation can slow innovation, but under-regulation can create safety and strategic vulnerabilities in a dual-use domain.
- CAUTION: The real success metric is not merely new entrants but whether the ecosystem can scale manufacturing, quality assurance and globally credible launch service delivery.
- CAUTION: The unfinished legal question: India regulates private space activity through executive policy (Indian Space Policy 2023) and

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.